Decoding Amplifies Bias: Measuring "Regard" in Open-Ended Text Generation

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Problem Statement & Motivation

Problem. Language models can produce different generations for prompts that differ only by a demographic mention (e.g., gender, religion, nationality). These differences can manifest as **social bias** in open-ended generation. We adopt the "**regard**" framing: how positively/negatively a generation *portrays* the demographic.

Open question we study. *Does the decoding algorithm (greedy / temperature / top-k / top-p) amplify or mitigate bias?* Even with a fixed model checkpoint, decoding changes diversity and repetition; it may also shift the **distribution of regard**.

Why it matters.

- Practical: decoding is a deployment-time knob (no retraining) that might reduce harmful outputs.
- Scientific: separates "model bias" from "sampling-induced bias".
- Feasible: end-to-end on Kaggle notebooks with controlled experiments and clear metrics.

Data

Data source(s).

- **Regard-labeled dataset & tools:** `ewsheng/nlg-bias` (`data/regard/*.tsv`; label in col1, text in col2).
- **Prompt templates:** paper-style templates (e.g., occupation/descriptors) with demographic slot; we will use a fixed prompt bank for reproducibility.
- **Generated samples:** GPT-2 completions per (prompt type × demographic × decoding).

License / usage rights.

- **GPT-2** and **BERT**: use released checkpoints under their published licenses/terms (cite in References).
- **nlg-bias repo:** if no explicit license is present, treat as research artifact: use with attribution; do not redistribute beyond class submission.

Preprocessing steps.

- **Mask demographics for scoring:** replace demographic mention with `XYZ` before running regard classifier (standard in nlg-bias workflow).
- Tokenize with BERT tokenizer; max length 128 (truncate/pad).
- Generation: fixed max new tokens; filter empty outputs; log prompts, seeds, and decoding config.

Baseline Plan

Baseline (first implementation).

- **Generator:** pretrained **GPT-2 small** (no fine-tuning).
- **Decoding: greedy** decoding only.
- **Bias scoring:** use **released regard classifier** from nlg-bias to label generations and compute group-level bias.

Core extension (main contribution).

- **Decoding sweep:** temperature/top-k/top-p (+ optional anti-repetition) and measure how **regard gaps** shift.
- **Classifier replication (optional but strong):** fine-tune a single BERT regard classifier on the released labeled set and compare to released model (accuracy + agreement).

Decoding grid (generation-time only).

Method	Rule	Knob(s)
Greedy	argmax token	–
Temp	sample softmax(z/T)	$T \in \{0.7, 1.0, 1.3\}$
Top-k	sample from top k	$k \in \{20, 50, 100\}$
Top-p	nucleus mass $\geq p$	$p \in \{0.8, 0.9, 0.95\}$
Anti-rep (opt.)	reduce repeats	no-repeat 3-gram

Fairness constraints.

- Same checkpoint, same prompt bank, same #samples, same max length across decoding settings.
- Multiple seeds per config; report variance.

Evaluation Plan

Primary bias metrics.

- **Regard distribution per group:** $P(\text{neg}), P(\text{neu}), P(\text{pos}), P(\text{other})$.
- **Negative-regard gap:** Δ_{neg} between groups within the same prompt type.
- **Uncertainty:** bootstrap confidence intervals over prompts/samples.

Generation quality controls (interpretation).

- Diversity: distinct-1/2 (unique n-grams / total).
- Degeneration: repeated n-gram rate; longest repetition span.
- (Optional) toxicity proxy score (aggregate only).

Protocol.

- Fixed prompt bank (30–80 prompts) × selected demographics (keep scope manageable).
- Samples per prompt/demographic: $N = 50$ (scale to Kaggle budget); 3 seeds per decoding config.
- Score masked text; run small ablation: masked vs unmasked scoring to test sensitivity.

Risks

Compute limitations.

- Kaggle runtime/quota limits may cap N or grid size.
- Mitigation: cache generations; prioritize high-signal configs (greedy, one temp, one top-k, one top-p).

Data quality concerns.

- Regard is subjective; classifier error can skew gaps.
- Mitigation: CIs + manual spot-check (20–40 examples) + classifier agreement if we fine-tune.

Potential failure modes.

- Bias shifts due to verbosity/length changes under decoding.
- Mitigation: fix max length; normalize by length; compare per-prompt rather than only global aggregates.

Ethics & Safety Considerations

Misuse potential.

- The study may surface offensive generations.
- Mitigation: do not publish large raw outputs; include only minimal excerpts needed for analysis + warnings.

Privacy / sensitivity.

- No personal data; prompts are synthetic templates; datasets are research artifacts.

Timeline (Week-by-week)

- **W1:** Prompt bank + GPT-2 runner; greedy baseline; caching/logging.
- **W2:** Integrate regard scoring; baseline bias gaps + sanity checks.
- **W3:** Decoding grid runs; compute bias + quality metrics; bootstrap CIs.
- **W4:** (Optional) fine-tune BERT regard classifier; compare agreement/robustness.
- **W5:** Ablations (masking, anti-repetition); finalize plots/tables + key findings.

What is happening inside?

Applicability. We do **not** fine-tune a very large generator; we use a small pretrained LM and (optionally) fine-tune a classifier.

Model architecture.

- **Generator:** GPT-2 (decoder-only Transformer, causal self-attention).
- **Regard classifier:** BERT-base (encoder-only Transformer) + linear head (4-way).

Training objective & details.

- Generator: no training; controlled decoding only.
- Classifier: cross-entropy; AdamW; early stopping on dev metric; report batch size/LR/epochs and best checkpoint.

References (2–6)

- Sheng et al. *The Woman Worked as a Babysitter: On Biases in Language Generation.* EMNLP 2019.
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